# IEEE P3335, Standard for Architecture and Interfaces for Time Card: Motivation and Progress Report

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***Abstract*—IEEE P3335 defines the generic architecture and interfaces of a Time Card: a modular timing subsystem that provides traceable time-of-day to a host system and, via protocols such as PTP and NTP, to networked distributed systems. The project generalizes the Open Compute Project (OCP) Time Card, the core of the Open Time Server, into a vendor-neutral standard. We report the motivation for the work and its progress as of mid-2026: a structurally complete, pre-ballot working draft covering conformance, architecture, performance declaration, timing and control interfaces, environment, and application guidance. The presentation summarizes the draft's key design decisions (a unified-timescale invariant, stackable conformance profiles, a transport-neutral control model, and declaration-based performance) and invites ISPCS review ahead of ballot.**

***Index Terms*—IEEE P3335, Time Card, IEEE 1588, PTP hardware clock, time synchronization, conformance, figures of merit**

## I. Motivation

Data centers, telecommunications, industrial control, finance, and scientific measurement increasingly depend on precise, traceable time-of-day, yet a typical host lacks an adequate local clock, clock steering, and time-transfer machinery. The OCP Time Card [1] (a PCIe card combining a GNSS receiver, a high-stability holdover oscillator, and an FPGA) showed that an open, modular timing subsystem can anchor an Open Time Server [2], and a diverse ecosystem of commercial and open implementations followed. Each uses product-specific signal definitions, control models, and performance claims: integration effort is duplicated for every card–host pairing, and independent comparison of products is impractical.

IEEE P3335 [3] (PAR approved December 2022, sponsored by the IEEE Instrumentation and Measurement Society) addresses this gap with three goals: plug-and-play interoperability between Time Card implementations and the hosts and operating systems that consume them, including POSIX integration via a shared PTP hardware clock (PHC); modular implementation around defined building blocks (time source, local oscillator, time processor); and unambiguous comparison through common figures of merit, measurement points, and operating conditions. The working group's guiding principle is to embrace implementation diversity while ensuring interoperability.

This abstract describes the unapproved IEEE P3335 draft; content is subject to change.

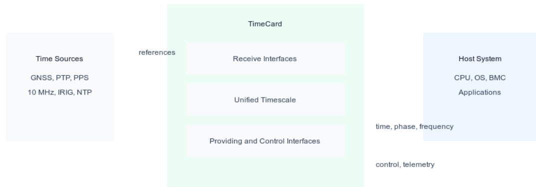


Fig. 1. Time Card context: external references enter through receive interfaces, discipline one unified timescale, and reach the host through providing and control interfaces.

## II. Architecture

The draft defines a Time Card by externally observable behavior rather than physical construction: one conformance model covers a discrete add-in card, embedded gateware, an FPGA or SoC function, and an external timing unit. Internally, the architecture names a local timing function (the oscillator state, from quartz to miniaturized atomic references), a time generator, a timestamping facility, and receive, providing, and control interfaces (Fig. 1); the timing data plane is separated from management and control.

Recognized receive interfaces include GNSS (multi-constellation, with antenna-state reporting), PTP per IEEE 1588-2019 [4] or IEEE 802.1AS-2025 [5], NTP [6], IRIG AM/DCLS, discrete 1PPS and frequency inputs, and other time-transfer methods such as White Rabbit and wireless time services. Receive-less designs (free-running or holdover-only) are permitted when declared, and multi-reference designs must document their selection policy (priority, validity, failure detection, fallback, recovery).

The central invariant is the unified timescale: a conforming instance generates exactly one, and every providing interface (1PPS, frequency outputs, time-of-day messages, PTP, NTP, IRIG, PCIe Precision Time Measurement (PTM), and host-bus timestamps) publishes, encodes, or derives from it, with a declared alignment error for each time- or phase-bearing interface. Ensemble operation must collapse to one selected or synthesized reference disciplining that timescale. Cross-output phase coherence thereby becomes a conformance property rather than a datasheet claim. Behavior at leap events, holdover entry and exit, reference failover, and manual steps must be documented, and continuous-phase outputs may not take undocumented phase steps while locked.

Host time transfer receives explicit treatment: mappings must document discovery, driver-visible registers or APIs, timestamp formats, and the latency, asymmetry, and correction data needed to evaluate time transfer into the host. PCIe PTM implementations must state which measurement point PTM timestamps represent and the correction terms relating PTM time to other providing interfaces, addressing the historically underspecified last segment between the timing card and the host operating system.

## III. Conformance and Performance Declaration

Base conformance requires at least one timing providing interface, one unified timescale, one accessible control interface, and full documentation of interfaces, performance metrics per measurement point, environmental limits, and validity conditions (warm-up, lock time, reference, temperature, measurement bandwidth). Conditional profiles stack on top: a Physical Timing Output profile with concrete 1PPS limits (SMA connector into a 50 $\Omega$ load, rising-edge on-time mark, 10%–90% rise time below 10 ns, alignment error below 100 ns between the 1PPS output and interfaces claiming alignment to it), a PCIe host-mapping profile, and Managed and Secure Infrastructure security profiles. The Managed profile requires authenticated, role-separated management with integrity-checked updates; Secure Infrastructure adds signed update payloads verified against a trust anchor, rollback protection, secure or measured boot before the timing service enters the locked state, and hardware-backed key storage. Claiming any optional feature binds the implementation to that feature's requirements.

Instead of universal numeric product classes, the draft mandates bounded, reproducible performance declarations. Each claim states its metric, measurement point, operating mode (locked, holdover, free-running, warm-up, failover, recovery), reference, environmental conditions, observation interval, uncertainty, and decision rule, with methods per IEEE 1139-2022 [7], IEEE 1193-2022 [8], and ITU-T G.810/G.8260 [9], and results metrologically traceable to a national metrology institute or a declared reference. MTIE is required for bounded time-error and holdover claims; TDEV, ADEV, phase noise, and a defined pulse-jitter statistic apply where relevant. Metrics may not be used interchangeably, and typical values cannot replace bounded ones. Informative annexes add formula-level metric definitions, a holdover error model, and example test procedures.

## IV. Control Interfaces

A transport-neutral baseline information model keeps object semantics identical whether control rides SMBus, I2C, or I3C, PCIe MMIO, IPMI or NC-SI, or REST, gRPC, Redfish, or SNMP. Required objects cover identity and versioning, device state (initializing through locked, holdover, degraded, and fault), timescale identity, sync-source status, alarms, security state, and coherent (tear-free) time reads. Conditional telemetry spans phase error in picoseconds, fractional frequency offset in parts in $10^{15}$, MTIE estimates, holdover elapsed time, and GNSS status. Configuration writes must report acceptance and whether they can cause a phase step, loss of lock, or output interruption.

## V. Progress, Process, and Remaining Work

The working group formed from the OCP time-appliances community after PAR approval; contributions on architecture, host integration, use cases, and metrics accumulated through 2025, and drafting moved to a public git repository [10] in October 2025. As of July 2026 the draft is structurally complete and pre-ballot: Clauses 1–10 plus metrics and test-procedure annexes compile to a 48-page document carrying 411 machine-indexed requirement statements. The draft is written as per-clause Markdown with pull-request-based comment resolution and a generated requirements index for traceability; the comment round from the April 2026 working-group meeting is merged, and all 14 normative references passed a technical use audit. Open pre-ballot decisions include ratifying the 1PPS electrical and alignment limits, choosing between a mandated cross-vendor host ABI (a common register layout or OS API) and transport-neutral semantics alone, freezing the baseline control vocabulary, ratifying normative-reference editions, fixing the security-profile cryptographic suites, and an independent timing-metrology review. The presentation will detail these decisions, share experience running an IEEE standards project on open git tooling, and solicit review and participation from the ISPCS community ahead of ballot.

## References


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